

Part 1: Mark the correct answer(s) for each of the following questions / statements. Please note that **multiple answers might be correct**.

- 1) Any set of digits or alphabets are generally referred as _____
 - a) Characters
 - b) Symbols
 - c) Bits
 - d) Bytes
- 2) The register that keeps track of the instructions in the program stored in memory is:
 - a) MAR Register
 - b) Program Counter
 - c) Current Instruction register
 - d) Status register
- 3) The currently executing process gives up the CPU voluntarily
 - a) In preemptive scheduling
 - b) In non- preemptive scheduling
 - c) All of the above
 - d) None of the above
- 4) What is the ready state of a process?
 - a) when process can be scheduled to run.
 - b) when process is unable to run until some task has been completed
 - c) when process is using the CPU
 - d) none of the mentioned
- 5) What is operating system?
 - a) collection of programs that manages hardware resources
 - b) system service provider to the application programs
 - c) link to interface the hardware and application programs
 - d) all of the mentioned

Part 2:

1. Fill out the following table

[6]

Decimal	Binary	Hexadecimal
48		
	0110 1101	
		2BE

2. Represent the following in 2s compliment

[8]

Integer	1 Byte representation	2 Byte representation
-14		
14		
30		
-34		

3. Represent the following in Signed Magnitude

[4]

Integer	1 Byte representation
55	
-55	
31	
-31	

4. What is the Maximum unsigned integer value that can be represented using 2 Bytes?

[2]

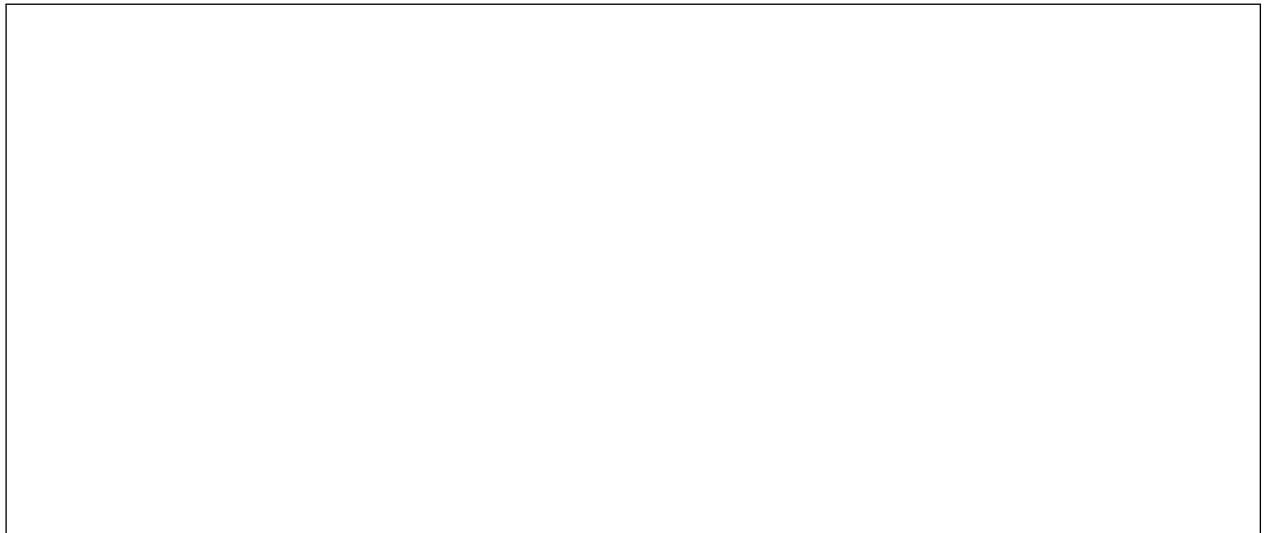
Part 3:

1. How many fetch-execute cycles will be performed if speed of processor is 700MHz. [2]

2. What is the purpose of base register and bound register? [2]

3. Draw the process state diagram, given the states of the processes. [5]

Ready, Waiting, Running, Terminated and New state



4. What is the purpose of DNS (domain name server)? [2]

5. Given the following memory map, fill the following table of memory map with jobs/processes numbers. Jobs/processes arrive as job1, job2, job3, job4 and job5 sequentially. [5+5]

Size	First Fit	Best Fit
100KB		
500KB		
200KB		
300KB		
600KB		

Memory Map

Job1	Job2	Job3	Job4	Job5
212KB	417KB	112KB	426KB	55KB

Jobs / Processes

Part 4:

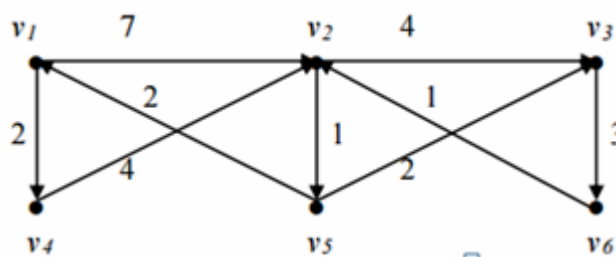
1. Differentiate between Vector and Bitmap graphics.

[2]

2. Differentiate between Digital Image Processing and Computer Graphics

[2]

3. Given the graph below, use method discussed in class to find the shortest path from the top left corner vertex/node (v_1) to the bottom right vertex/node (v_6). Write number for all the vertices/nodes accordingly. Give the length of the shortest path.

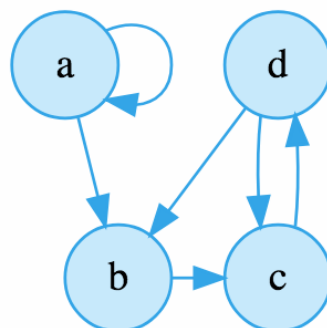


[6+2]

4. Differentiate between supervised and unsupervised learning with an example. [1+1+1]

5. Differentiate between classification and regression with an example. [1+1+1]

6. Create adjacency matrix for the following graph. Assume that each link has weight 1. [4]



7.